

Current Mirror: Simple and Cascode

Student	Deva Prassad S
Roll Number	2025BEC0019
Programme	B.Tech Electronics and Communication Engineering
Institution	IIIT Kottayam
Tools	Cadence Virtuoso

1. Aim

To design and simulate a simple current mirror and a cascode current mirror in Cadence Virtuoso, to obtain 1mA and 2mA currents.

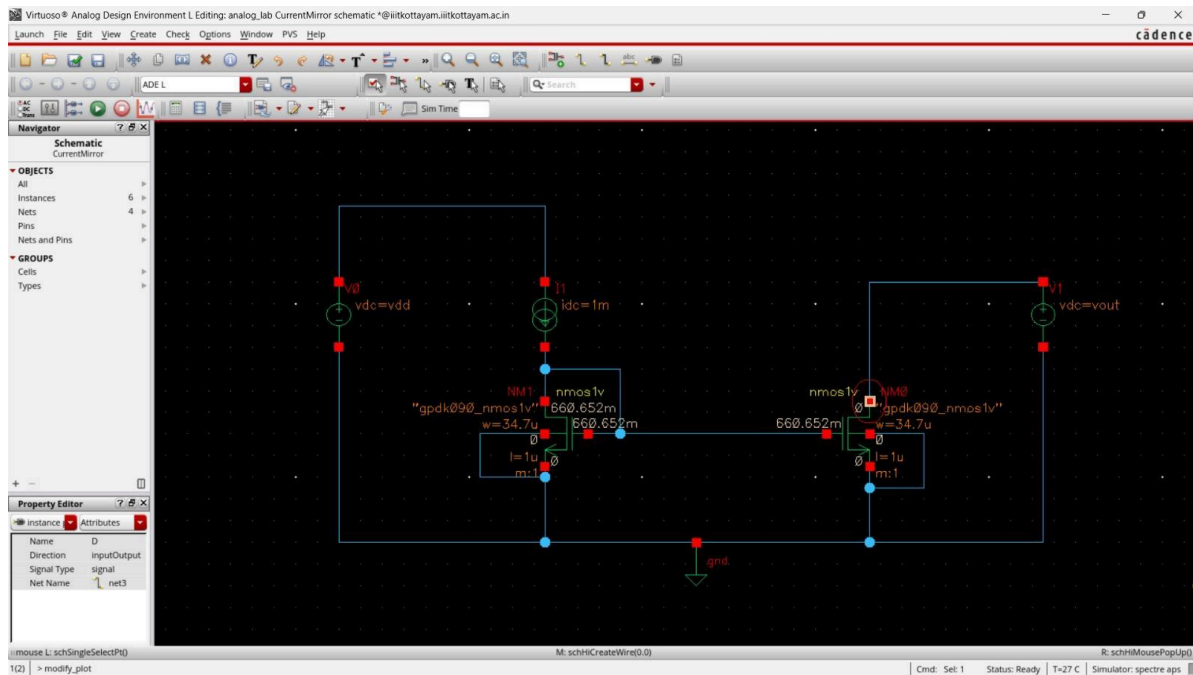
2. Design Specifications

Variant	W _{ref}	W _{out}	L	I _{ref}	Expected I _{out}
Simple (1mA)	34.7 μm	34.7 μm	1 μm	1 mA	$\approx 1 \text{ mA}$
Simple (2mA)	34.7 μm	69.4 μm	1 μm	1 mA	$\approx 2 \text{ mA}$
Cascode (1mA)	34.7 μm	34.7 μm	1 μm	1 mA	$\approx 1 \text{ mA}$
Cascode (2mA)	34.7 μm	69.4 μm	1 μm	1 mA	$\approx 2 \text{ mA}$

3. Circuit

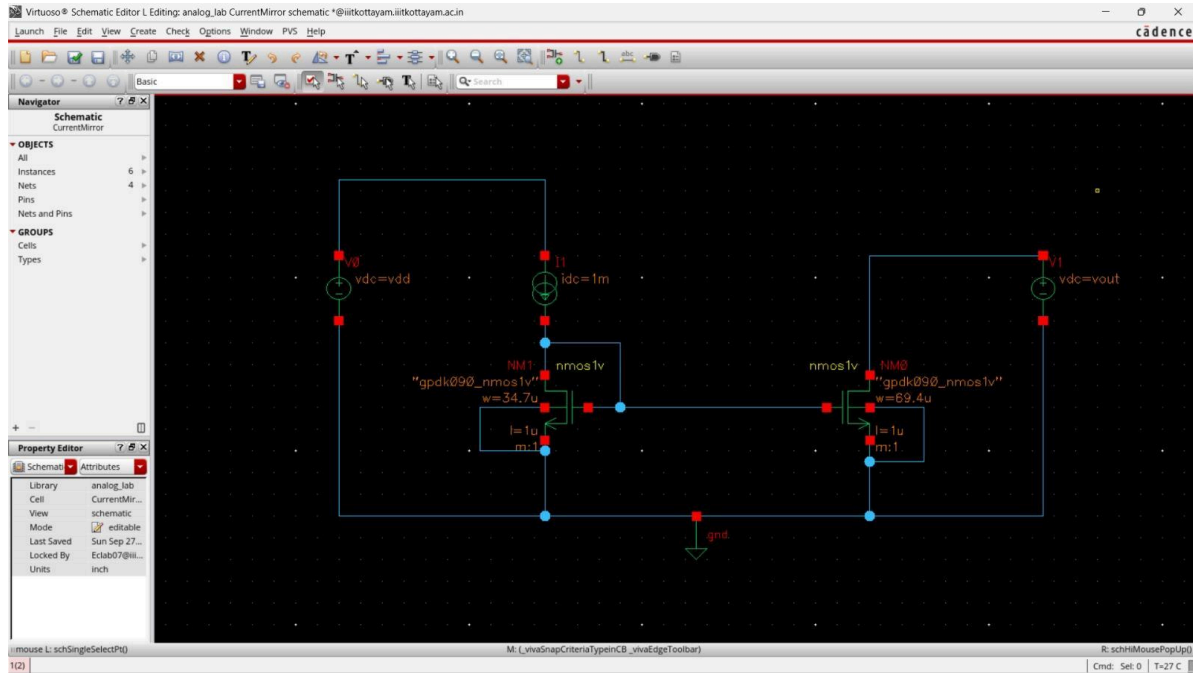
3.1 Simple Current Mirror (1mA)

The reference device (diode-connected, gate tied to drain) and the output device are both sized at $34.7\text{ }\mu\text{m} / 1\text{ }\mu\text{m}$. A channel length of $1\text{ }\mu\text{m}$ — well above the process minimum — is used on both devices so that channel-length modulation stays small and each transistor's intrinsic output resistance $r_o = 1/(\lambda \cdot I_D)$ remains reasonably high; this gives a fairer, more visible baseline against which the cascode mirror's extra output resistance can be judged later. The reference width of $34.7\text{ }\mu\text{m}$ places the diode-connected device at a practical saturation overdrive for the 1 mA reference current — wide enough that its V_{GS} doesn't consume excessive headroom, but not so wide that device area or capacitance is wasted. With the output device matched to the reference width, the mirror ratio $W_{out}/W_{ref} = 1$, so the output current is expected to track the 1 mA reference current directly.



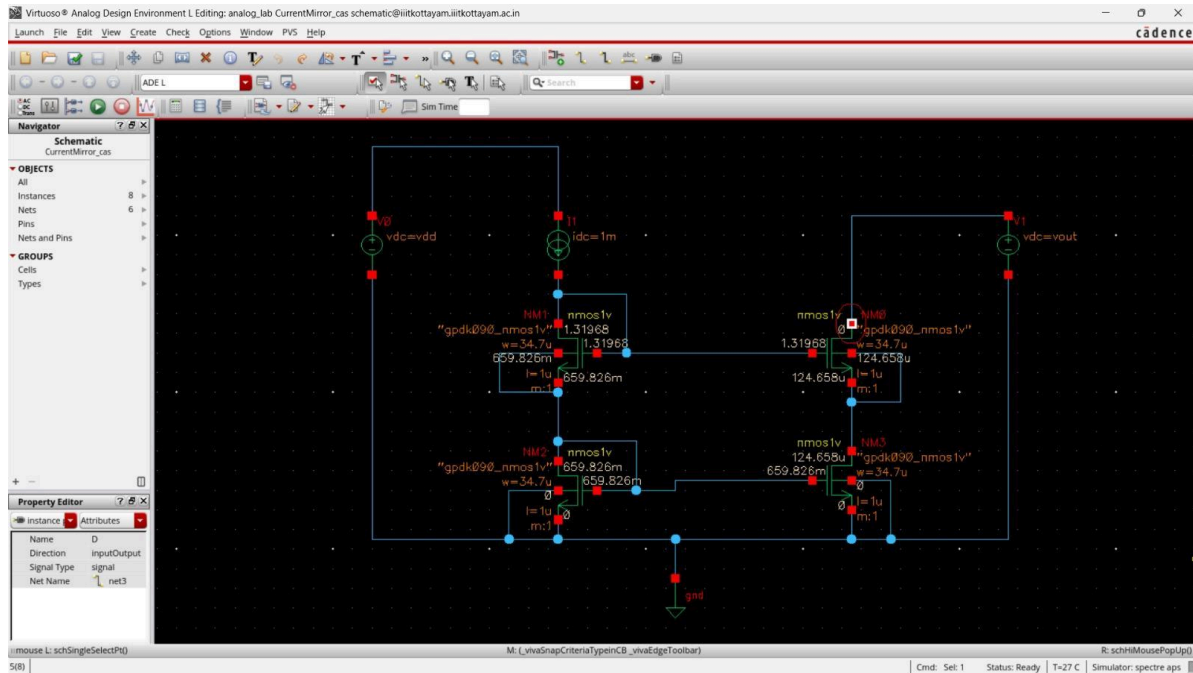
3.2 Simple Current Mirror (2mA)

The reference device is kept at $34.7 \mu\text{m} / 1 \mu\text{m}$, but the output device width is doubled to $69.4 \mu\text{m}$. Since both devices share the same VGS (set by the diode-connected reference) and the same channel length, doubling the output width doubles its effective W/L, so by the square-law current-mirror relation $I_{\text{out}}/I_{\text{ref}} = W_{\text{out}}/W_{\text{ref}}$, the output current should scale to approximately 2 mA.



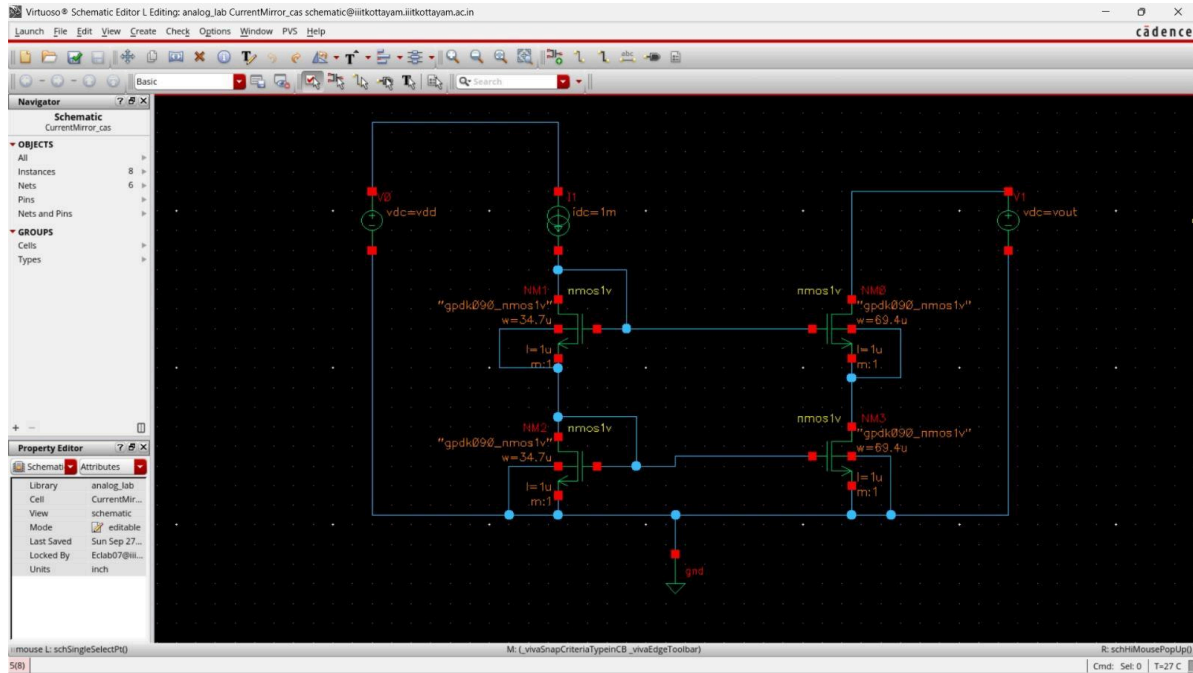
3.3 Cascode Current Mirror (2mA)

A second diode-connected device is stacked above each transistor of the simple mirror, and every device in the stack — reference pair and output pair — is sized identically at $34.7\text{ }\mu\text{m} / 1\text{ }\mu\text{m}$. Keeping the cascode device the same width as the transistor beneath it in the same branch means both devices carry the same current at the same overdrive voltage, so the cascode addition does not disturb the current ratio already fixed by the lower mirror pair. What it does add is a series gm-ro boost, raising the resistance looking into the output node well above that of the simple mirror, at the cost of needing extra VDS (VOV of two stacked devices) before the output settles.



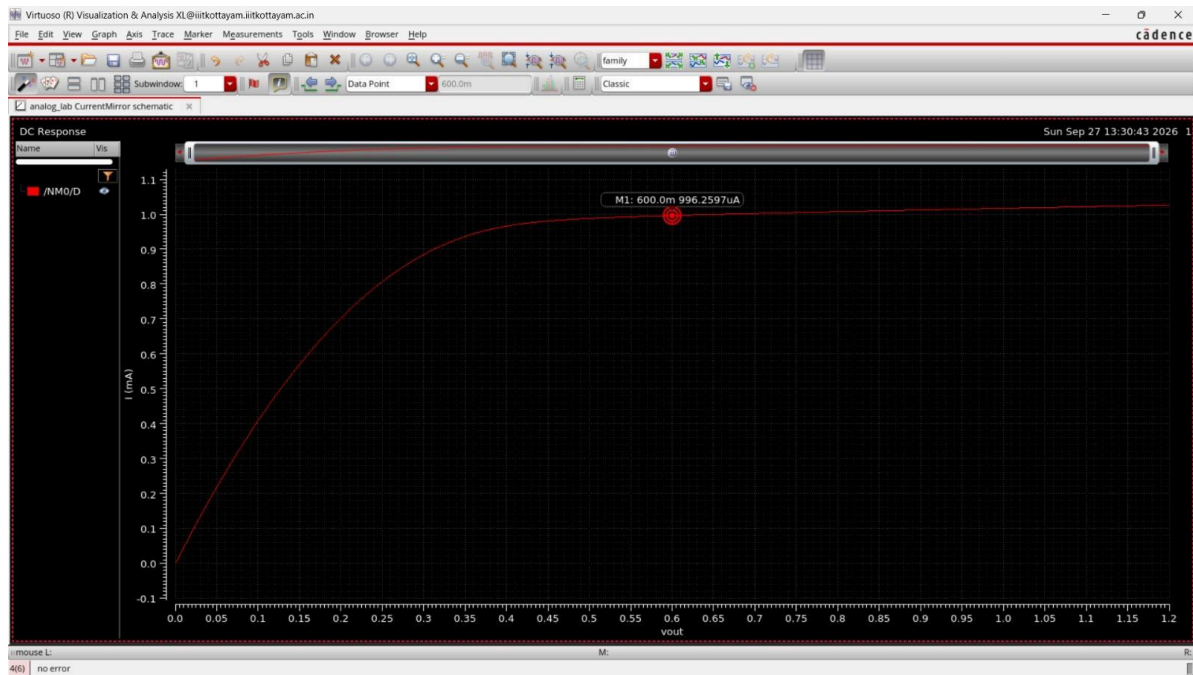
3.4 Cascode Current Mirror (2mA)

The reference stack (both diode-connected devices) is kept at $34.7\ \mu\text{m} / 1\ \mu\text{m}$, while the entire output stack — both the output mirror device and the cascode device above it — is widened to $69.4\ \mu\text{m}$. Widening both devices in the output branch together, rather than just one, preserves the same overdrive voltage up and down that branch, so the 2:1 width ratio between the output and reference branches sets the mirror gain ($I_{\text{out}} \approx 2 \times I_{\text{ref}}$).

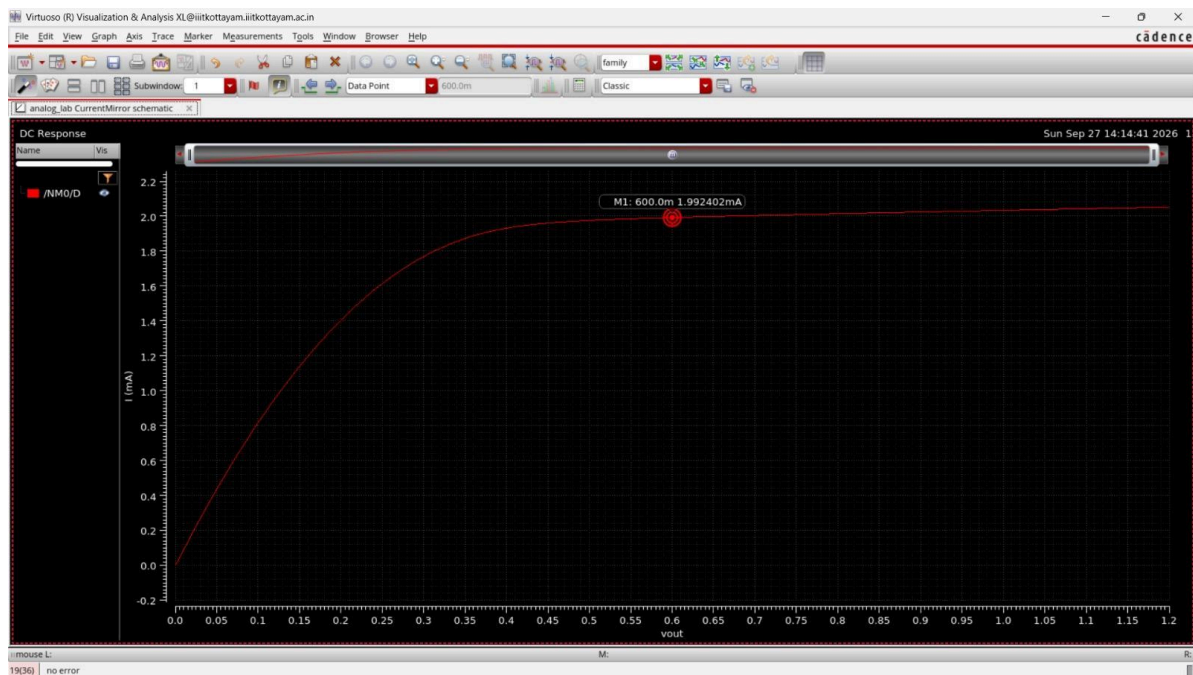


4. Results

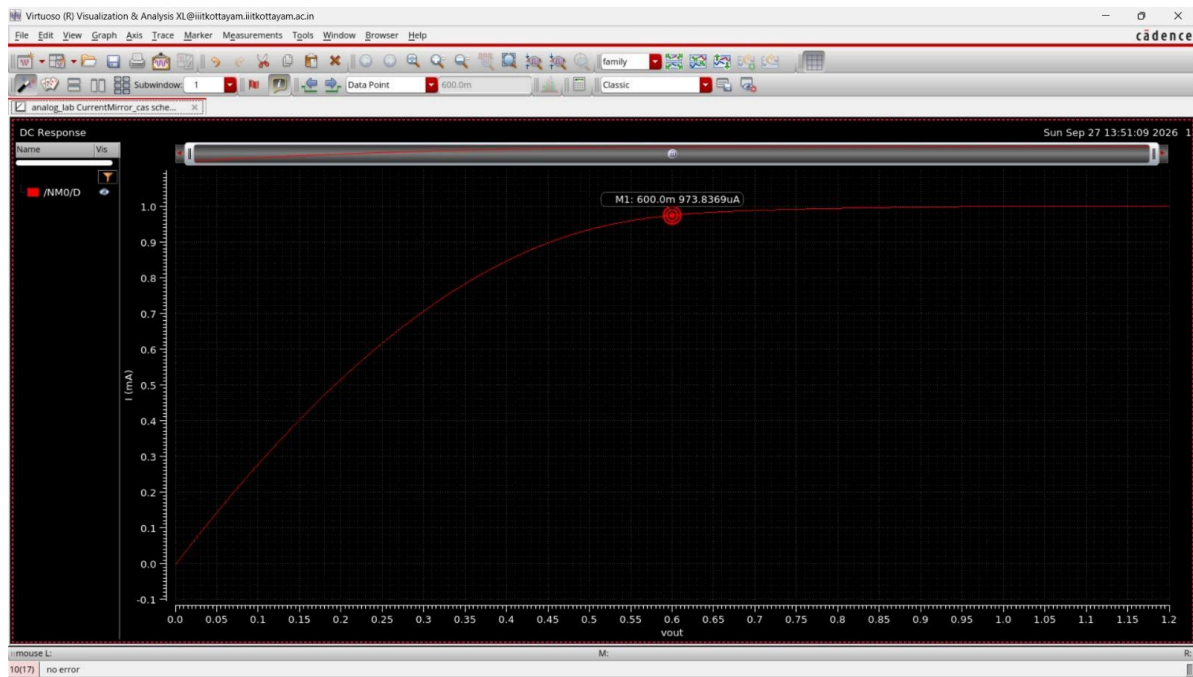
4.1 Simple Current Mirror (1mA)



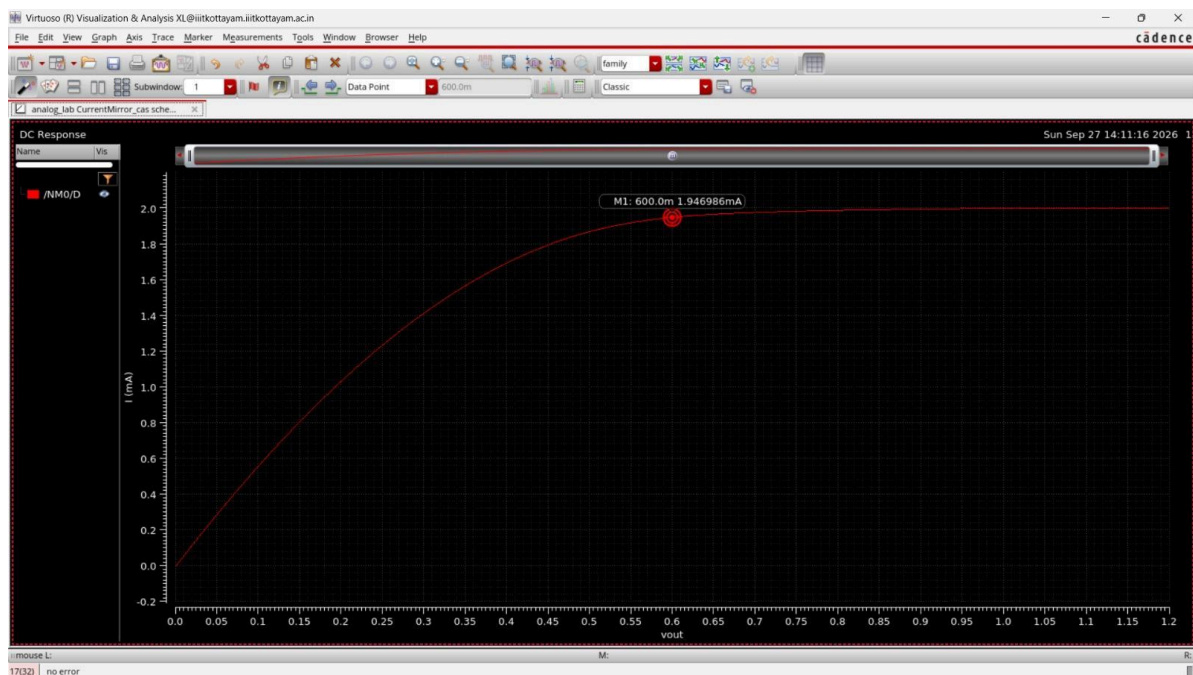
4.2 Simple Current Mirror (2mA)



4.3 Cascode Current Mirror (1mA)



4.4 Cascode Current Mirror (2mA)



5. Observations

- Both topologies reproduce their reference current (or twice for double the reference width) once the output device enters saturation, confirming $I_{out}/I_{ref} = W_{out}/W_{ref}$ holds for the simple mirror as well as the cascode mirror.
- The simple mirror's output current keeps rising gradually across the swept output voltage, indicating a comparatively low output resistance, as expected from its shorter effective r_o path.
- The cascode mirror's output current flattens out and stays essentially constant for the rest of the sweep, confirming the higher output resistance gained from the stacked diode-connected devices.
- The cascode mirror needs a higher output voltage before its current settles, compared to the simple mirror.

6. Result

Simple and cascode NMOS current mirrors were designed and simulated in Cadence Virtuoso at two width ratios (1:1 and 2:1). The device widths and channel length were chosen to give the reference device a practical saturation overdrive while keeping channel-length modulation low, and the output/cascode widths were set to realise the required mirror ratio while preserving matched overdrive voltages within each branch. Both topologies accurately reproduced the current scaling expected from their width ratios, and the cascode configuration showed a flatter output characteristic and higher output resistance at the cost of a higher compliance voltage.

7. Inference

Cascode current mirror is used in cases where high output impedance and high precision current is required.

Simple current mirror is used in places where the input voltage is low.